

Davi Guanabara Aragão

AI/ML Engineer | PhD Researcher in Autonomous Systems and Data Science

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PROFESSIONAL SUMMARY

AI/ML Engineer and PhD researcher at the Aeronautics Institute of Technology (ITA), with experience applying artificial intelligence, machine learning, and data science to simulation-based studies in aerospace, autonomous systems, geospatial analysis, and environmental prediction. Develops and evaluates computational models, experimental pipelines, and research software for decision-making, multi-agent coordination, and data-driven analysis in complex engineering contexts.

AREAS OF EXPERTISE

Autonomous Systems: Autonomous Decision-Making • Multi-Agent Systems • Autonomous UAV Systems

Machine Learning: Deep Learning • Reinforcement Learning • Attention Mechanisms

Software Engineering: Modular Software Architecture • Event-Driven Systems • End-to-End AI Systems • Experimentation Frameworks

Geospatial Computing: Geospatial Modeling • Geo-Temporal Data Analysis • Spatial Intelligence

SELECTED PROJECTS

Deep Reinforcement Learning with Graph Attention for Cooperative Loyal Wingmen (2024--2026)

Journal of Intelligent & Robotic Systems (Springer)

- Developed a deep reinforcement learning framework for cooperative threat-engagement decision-making in Loyal Wingmen UAVs operating under shared spatio-temporal observations.
- Proposed a graph-attention policy architecture enabling adaptive information sharing and coordinated decision-making under partial observability.
- Designed and evaluated simulation-based experiments integrating graph neural networks and multi-agent reinforcement learning for autonomous aerial systems.
- Published in the *Journal of Intelligent & Robotic Systems*. DOI: 10.1007/s10846-026-02415-8.

SkyWeaver --- Dynamic Airspace Configuration Platform for UTM/U-Space (2025--Present)

Research Platform

- Architecting a research platform for Dynamic Airspace Configuration, enabling the design, evaluation, and comparison of operational airspace scenarios for future UTM and U-Space ecosystems.
- Developing a reactive software runtime based on event-driven architecture, supporting dependency-aware execution, operational state management, and digital-twin-oriented applications.
- Investigating graph-based optimization methods for droneport placement, airspace infrastructure planning, and route optimization under operational constraints.
- Exploring learning-based graph optimization techniques to support AI-assisted airspace topology generation and operational planning.
- Research vision integrates software engineering, optimization, and artificial intelligence into decision-support tools for future Advanced Air Mobility (AAM) operations.

OndeVale --- End-to-End Geospatial Property Valuation Platform (2025)

Cloud-Deployed AI System

- Developed a cloud-deployed platform for residential property valuation combining geospatial analytics, machine learning, and interactive web visualization.
- Designed the complete machine learning pipeline, including data preprocessing, feature engineering, model training, inference services, and deployment.
- Built a FastAPI backend and Vue.js frontend integrating GIS visualization with real-time model inference.
- Deployed the application as a publicly accessible cloud service, delivering an end-to-end AI solution from data acquisition to user interaction.

Deep Learning for Geo-Temporal Environmental Modeling (2024--2025)

- Developed deep learning models for vineyard leaf wetness prediction using meteorological observations from the Iowa Environmental Mesonet.
- Designed a Deep Sets architecture with attention mechanisms to learn permutation-invariant representations from geo-temporal environmental data.
- Built a reproducible experimentation framework integrating Bayesian hyperparameter optimization, bootstrap evaluation, and deterministic training workflows.
- Investigated attention-based neural architectures for environmental prediction using heterogeneous meteorological observations.

EDUCATION

Ph.D. in Electronic Engineering and Computing (2025--Present)

Aeronautics Institute of Tecnology (ITA)

Graduate Certificate in Data Science (2024--2026)

Aeronautics Institute of Tecnology (ITA)

M.S. in Aeronautical Computing (2021--2024)

Aeronautics Institute of Tecnology(ITA)

B.S. in Control and Automation Engineering (2013--2019)

University of Fortaleza (UNIFOR)

SELECTED PUBLICATION

de Aragão, D. G., Maximo, M. R. O. A., & Brancalion, J. F. B. "Deep Reinforcement Learning with Graph Attention for Cooperative Loyal Wingmen under Shared Spatio-Temporal Observations." *Journal of Intelligent & Robotic Systems*, 2026.
DOI: 10.1007/s10846-026-02415-8